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## On the Estonian Internet Voting System, IVXV, SoK and Suggestions

**Abstract.** The Estonian i-voting experience is probably the richest to analyze; a country that is considered a pioneer in digitizing both the government and private sector since 2001 followed by online internet voting (i-voting) in 2005. However, there are still some complaints submitted, critics and remarks to consider about the IVXV system. In this paper, we introduce a Systemization of Knowledge of the Estonian IVXV i-voting system and propose some added security enhancements. The presented SoK discusses applications implemented by election observers, which, to our knowledge, have never been mentioned and/or analyzed in the academia before. We also identify and analyze recent fixes and improvements in the June 2024 version used in the European Parliament elections connecting them to their academic sources, and grasp what we could trace in the 2025 version released while we write. In addition, we point out **to two** unnoticed automated formal verification **attempts** of IVXV; the researchers discovered a privacy attack that we show extendable to a possible large scale encrypted vote copying. Finally, we discuss the current system status and the ongoing efforts towards providing post quantum solutions.

**Keywords:** IVXV, El-Gamal Encryption, universal verifiability, End-to-End Verifiability (EEV), counted-as-casted, vote buying/coercion.

### 1 Introduction

Estonia is a small 1.35 million population country located in east Europe who gained independence from the Soviet Union in 1991 and joined the European union in 2004 [1].<sup>1</sup> Most Estonian citizens welcomed the earlier *digital transition*; however, when it came to e-voting in 2005 there were some kind of “*notable divisions within the society between those who fully trust and those who fully distrust internet voting*” as quoted from the **OSCE-ODIHR** (*Organization for Security and Cooperation in Europe- Democratic Institutions and Human Rights*) **2023** report [2]. This is reflected clearly in the i-voting statistics; although the Estonian experience could be viewed as the earliest and most advanced [3], the official site [4] shows the ratio of i-votes reaching its maximum of 51% in 2023 (local Parliament), down to 41.7% in 2024 (European Parliament), then **little up again to 45.8% in the just ended 2025 local elections**. One can trace a long history of objection incidences mostly from right wing parties in [1] and [2/(page 8, footnotes 16&17)]; the situation was emphasized in 2023 when internet votes flipped the results for one of those parties (**EKRE**). Analysts view

<sup>1</sup> The live number in 30/4/2025 is 1,347,056 (<https://www.worldometers.info/world-population/estonia-population/>). More detailed statistics, but dated to 8/2024, are in (<https://www.stat.ee/en/find-statistics/statistics-theme/population/population-figure#>); **1,127,312 “citizens”** :296,268 ~ **26.28% are ethnic Russians**, (~ 1.35m-1.127m) are ≥ 1 yr residents.

it as a natural echo of the society division mentioned above; i.e., it is expected for curves, [5], showing the distribution of internet votes to be completely different from paper votes curves according to each party's preference. Still, there were some complaining activities that continued persisting to the 2024 European elections [6] and the current 2025 local elections [7]<sup>2</sup> going as we right.

To complete the picture, it is appropriate to note that Estonia shares a border with Russia (recall that ~26% citizens are ethnic Russians<sup>1</sup>); according to [8] the ongoing war has put extra pressure on the country. The i-voting<sup>3</sup> system faced Russian attacks that authorities say were properly defended. Since most security metrics are probabilistic [9/sec. 4], the power and incentives of possible attackers have an impact on the probability of an attack to succeed; whether it is just a party competing for a win, or a powerful country that has farsighted interests. However, we here just notify of a possible impact and leave the analysis of measuring/estimating it to political science researchers.

After this brief preface on the Estonian election environment and the involved players, we proceed into the technical and cryptographic details; hence, the rest of the paper is organized as follows. Section 2 reports some recent important activities by the i-voting opposing community that have technical merit, while section 3 marks briefly the milestone steps through the evolution of the Estonian i-voting system<sup>4</sup>. Then section 4 explains in detail the current version of IVXV ending with an important attack that was fixed before the 2023 elections, while section 5 goes through recent improvements that were made in IVXV before the European elections and section 6 scans briefly the updates we recognized in the version just released on the 3<sup>rd</sup> of October. Section 7 discusses the remaining vulnerabilities of the system along with suggested solutions that address the concerns of EKRE as stated in the OSCE 2025 report [10/sec.3.3/28] and ends by discussing the quantum computing threat and efforts toward it. Finally, section 8 introduces suggestions for further future research and concludes the paper, while Appendix A introduces summarizing tables.

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<sup>2</sup> Prof. AGO Samoson is an NMR & materials science researcher in Tallin University of Technology (TUT) (<https://www.researchgate.net/profile/Ago-Samoson>); as IVXV representative clarified, "Cybernetics is NOT the same institution as Cybernetica, it shares some common history, but this ended more than 25 years ago and the researcher Ago Samoson is in no way affiliated with Cybernetica, nor IVXV development", where Cybernetica is the company behind Estonian i-voting system partnering with Smartmatic. However, we hereby witness that although he is against i-voting, he informed the authors about new improvements and the research papers behind them (section 5.2).

<sup>3</sup> The term "*i-voting*" refers to voting through the internet (online) which is a subset from the more general term "*e-voting*" that refers to all kinds of electronic voting that may include using Ballot Marking Devices (BMDs) at poll stations; hence, both terms can be used to describe the Estonian voting system.

<sup>4</sup> Since there were many previous studies discussing earlier years of the Estonian i-voting, like [1] till 2019, this paper is more concerned about the vulnerabilities and improvements of the recent years. Our title says IVXV; a design that took place in 2017.

## 2 Recent Opposition Activities with Technical Merit

Discussing the Estonian environment and exploring possible players in its political game with their variant power resources, is an illustrative example of how digital democracy enforced security researchers to investigate the geo-political prospective. Vice versa, this section illustrates how electronic voting partially turned the opposition-system classical debates about election results<sup>5</sup> into technical ones. We find observers who read papers and acquire programming skills, while political parties are suspicious of what they do not know and must learn; finally, courts are cautious and sometimes confused by experts. Section 2.1 presents activities and complaints that accompanied the 2023 local elections, while section 2.2 covers 2024 European elections, then section 2.3 goes through the currently ongoing activities of the 2025 elections that just finished while writing this version of the paper.

### 2.1 The 2023 Local Elections

A technical incident that gained some publicity in 2023 elections [11] was done by the same computer scientist observer<sup>6</sup> in [5]; Mart Põder *voted using his own Python code* [12,13]. This highlighted the known fact that the voting application, which voters should download to deliver their vote, is not authenticated by the system; the OSCE report [2/page 8] believes the incident “*could present a cyber security risk*”. The report also mentions *some wrong district votes*; the incident was magnified by the opposition [14,15], while described by IVXV representatives [9/ sec. 5] as a population registry problem that would have taken much longer to detect and resolve without i-voting.

Even if some exploits and/or problems were magnified, it is the authors’ impression that most complaints get rejected without objective investigation based on a submission deadline (*3-days from election*); [16] is a very recent (20/6/2025) example. The OSCE 2025 report comments on this issue by saying [10/sec. 3.3/68 & Recomm. J] “*the legislator should also address the specific complexities of internet voting when determining appropriate deadlines*”. To be fair we noticed that in many cases, as will be illustrated throughout the paper, the vulnerability gets handled and fixed silently before the following election.

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<sup>5</sup> Everywhere all the time, losers always complain and winners always defend; one of the hardest jobs is to extract the facts and the objective arguments from the equally motivated sides of the story.

<sup>6</sup> The word “*observer*” is used by IVXV to acquire certain access rights during election (as opposed to “auditors”). The term “*Computer Scientist*” (taken from [9]) is rejected by IVXV representatives who describe Märt Põder as “*A hobby hacker activist*”, while the OSCE report referred to who codes another voting application as “*someone with sufficient programming skills*”.

## 2.2 The 2024 European Parliament Elections

We start by another rejected complaint about the decryption of invalid votes (after the elections), where the rejection was done objectively this time [17]. Among the three listed reasons, being an *observer* not an *auditor* seems to be the dominant one, where auditing is organized by the State Electoral Office in all elections. According to [18/Conclusion], generating proofs of correct decryption of invalid votes was remediated in code before June 2024, but the file containing the decryption of invalid votes is only accessible to auditors [18/page 22]; the reasons will be further discussed in section 5.1. The version released in October 2025 for the local elections prevents invalid votes from entering the system at all; however, this does not apply to other unofficial voting application voters may use as will be discussed further in section 6.

In addition to the complaint in [16,17], and the rejecting voices [19], the European parliament elections were accompanied by some activities from the i-voting opposing community. The same observer mentioned above has developed some kind of shadow e-voting site called *virtual threshold survey* [20] encouraging citizens to vote again on it as a check (although there is no evidence of considerable participation ratio).

## 2.3 The 2025 Local Elections

In the current 2025 local elections, observers have prepared a verification application, [21], that reveals the vote decryption and had to update their 2023 voter application<sup>7</sup> code to handle the new ballot format in the new version; they are broadcasting a tutorial video, [22], to show voters how to use it. In the course of doing so, they filed a complaint to the National Electoral Committee, on October the 3<sup>rd</sup>, for the delay in publishing the system code, only at 30<sup>th</sup> of September, which does not give them enough time to study and analyze the updates; after the first rejection, the appeal to the supreme court upheld the complaint [23,24] and declared it “*unlawful that the State Electoral Office (RVT) failed to disclose the source code of the electronic voting system and the verification application prior to the test vote*”.

Another circulating issues in 2025 elections are not publishing the voters’ list [25], the handling of the parts of the election secret key (during the setup phase [26,27]), and some incidents, [28], that happened during the re-reading and integrity checking phase. The EKRE party submitted several complaints, [29], from the Electoral Commission to the supreme court that we will discuss them as we go.

### *The key parts*

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<sup>7</sup> The terms “Voting Application” and “Voter Application” both abbreviated as VA, are used interchangeably in available literature, documentation and news coverage to refer to the application that runs on the voter’s device to receive the voter’s choice, turn it to an encrypted digitally signed ballot, and sends it to the system.

-According to EKRE submitted complaint [26] and a supporting observer [27], pin codes protection was removed from the chips containing the key parts this year<sup>8</sup>, while [27] claims the chips were kept with the key shareholders outside the meeting room with them swearing they did not misuse it. The first complaint was rejected on the basis that it is technically impossible to read and/or copy isolated chip cards from a distance<sup>9</sup>.

-It is worth mentioning in this context that there was also an older complaint about the desktop used in keys creation not being completely isolated and having some extra software downloaded onto it. A first earlier complaint granted an observer (23/2/2023) permission to see the content of the backup copy of the boot hard disk used in key creation to have full confidence there is no malware in the computer memory during key creation [30]. The observer took the photo shown in Fig.1 when the disk inspection took place (28/11/2023), then pursued the matter further to the supreme court. The supreme court responded that this could not have affected the voting results; quoting their exact words, [31], "*voting results cannot be compromised with malware, because with the help of the reading certificate issued when determining the voting results, the compromise would be revealed immediately*".<sup>10</sup> On Aug 2024, a commentator from IVXV team stated (in an earlier reviewers' report of an earlier version of this paper) that they admit the risk and "*it has been taken care of*".



**Fig.1.** An image taken from [30]. According to the observer, this is the computer used in key creation which was supposed to have only an authentic Windows10 operating system, but he detected other software installed on it; DigiDoc4, Notepad++ and RamDisk.

<sup>8</sup> It could be, we anticipate, that EKRE refers to the new eID company, Thales, mentioned in footnote 14 that does not use pin-code authentication; however, the available documents state that it will start on 16<sup>th</sup> Nov, i.e. not used in this election.

<sup>9</sup> First, we notify that revealing the election secret key alone can only lead to a privacy risk (revealing how voters voted), unless it was accompanied by a malicious voting application and non-verifying (or absent) voters; this will be explained further in the following sections. As for reading chip cards, one can map at first thought to Bitcoin hard wallets where guarantees are made for people money must include passwords in addition to isolation; in addition, if [27] claims were true reading from a distance is not the only possible risk. **A reader interested in election key management of different e-voting systems can check this recent, EVoteID25, SoK** ([https://link.springer.com/chapter/10.1007/978-3-032-05036-6\\_2](https://link.springer.com/chapter/10.1007/978-3-032-05036-6_2))

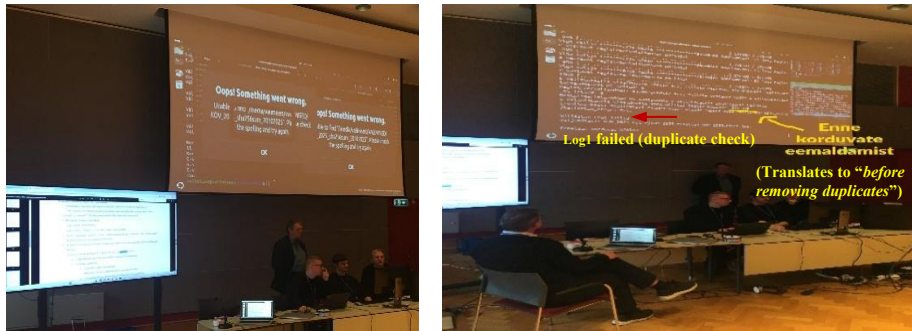
<sup>10</sup>The official progress of events is in (<https://www.riigikohus.ee/et/laheidid/?asjaNr=5-23-40/2>)

### *The re-reading incident*

During the re-reading and integrity checking of the 2025 election results, [32/mins 00:45, 15:50] and Fig.2, two issues were reported by observers [28, 33]. The first is the system failing to read the checksum of a vote, and the second is an integrity alarm declaring a duplicate vote; the voter’s information was disclosed due to the errors. This was included in the rejected EKRE complaint mentioned above, [26], as one of the reasons to doubt the whole process. The election authority on the other hand released a statement, [34], described the circulating video as “*misleading*” and clarifying that the system reading error was due to a vote coming from a different voting application which did not accurately comply with the new ballot format, while the alarm came from a voter’s error in repeating the same vote twice which does not reflect a problem in the system. The EKRE sub complaint about the whole process was dismissed by the Electoral Commission [35].

We, the authors of this paper, are convinced by the official justification of the first issue; the complaint in [23,24] partially reflects the tedious format update issue discussed by the same person in [21 or 22]. However, the EKRE posts, [26], comments on the fact that the voter (his identity was revealed) unfairly lost his vote; the system did not send him any feedback error message<sup>11</sup> to revote.

We disagree though with the argument explaining the duplicate vote alarm; it should not be a problem if the same voter revoted with the same exact choice again, the added randomness (see section 4) makes the two votes encrypt to completely different cipher texts. We believe if the system here detected two duplicate ciphers, this *could possibly reflect a malicious voting application that submits the same ballot no matter what the voter chooses*. **We recommend adding a general check of duplicate ciphers exactly after removing voters’ signatures and while they are still available to trace**; this will be further discussed in section. Also, since the voter’s ID was already revealed during the process, we suggest checking his side of the story on the exact details of what he did.



<sup>11</sup> Apparently, their voting application delivered the ballot directly to the vote collector application, while the official one has a server-side part (<https://github.com/valimised/ivxv/blob/published/voting/service/voting/main.go>) that performs some checks and sends error messages.

(2-a) min 00:45 not able to read checksum (2-b) min 15:51 log1 check failed due to duplicate  
**Fig.2.** Snapshots taken from [32] before blurred as shown in [28] (the red arrow and the English caption are added by the authors of this paper)

### ***The published code vs the used one***

Commenting on the doubts raised by EKRE in [29,35] about “*how can be sure the published code is the one they actually run?*”; this can be achieved through file digests or fingerprints. The cryptographic hash of the published code that is claimed to run in the re-reading process can be calculated ahead of the meeting, then calculated again during the meeting before they start the run. This can be done also to the voting application code if it is ever published.

### ***Absent e-voting***

It is also worth mentioning that some voters, [36], say that when voting at poll stations, they were told that this is not their first vote; the system records an earlier electronic vote for them. It is true that a poll station vote cancels the electronic vote; i.e., in this specific case, the voter’s choice was recorded correctly. However, accumulating with the previous incident, we believe this deserves an investigation; retrieving all available details of the electronic vote to know how this exactly happened and whether there is a non-negligible probability it was repeated; extreme possibilities like those in [37] will be further discussed in section 7. **In this context, we notify that election participation ratio (whatever through the internet or in poll stations) usually ranges between 50-60% except for the European elections 2024 was down to 37.6% participation ratio.**

Finally, another scientific report from *the Cyber Security Committee of the Academy of Sciences* has been handed to the election organizers [38]. The complete report is not published, but the minutes of the committee meeting [39] on 3<sup>rd</sup> of June 2024 clarify they have identified 6 threats whose risk class is higher than small<sup>12</sup>. We chose to mention the report here, although it is not an opposition activity, to illustrate that there are some problems; i.e., and not all opposition complaints are exaggeration. However, to be fair, we should stress that this report was before the improvements and/or fixes made in 2025.

## **3 Earlier System Evolution**

As mentioned earlier, digitization has been in Estonia for more than 20 years, even before 2001, and was extended to include the private sector hand in hand with the e-government; e-ID cards existed since 2002, and electronic transactions is the casual behavior of the Estonian citizen. While details on digital system architecture and components like *Xroad*, *KSI* private blockchain seem irrelevant here, we find the e-ID key generation relevant since it is used in internet voting from its beginning in 2005.

<sup>12</sup> A system representative commented (14/8/2024) that all the 6 threats are of risk class medium (11-13); we have no evidence on that except a previous review report of this paper.

Hence, we will dedicate section 3.1 to one major event that changed a core cryptographic component of the e-ID system, **RSA**; then we will follow with a brief overview of i-voting earlier evolution till it reached its main design as IVXV in 2017.

### 3.1 Electronic Identity Card 2018 problem

In May 2018, Estonian authorities officially declared a persisting problem that started to appear in some rare incidences of duplicate RSA keys since 2011/2012. Citizens with problematic keys were asked to re-install the Java Applet on their cards at PPA (Police and Border Guard Board = *Politsei-ja Piirivalveamet* in Estonian language) stations. When reinstallation became more frequent, researchers started to analyze the accumulating data for the root problem. Then, it was proven that the ID card manufacturing company, **Gemalto**, generated the RSA keys outside the chip, which violates the agreement rules and gives a chance to copy and/or repeat key pairs. A lot of interesting details on how the analysis was done can be found in [40]; more faulty keys issues<sup>13</sup> can be found in the same researcher's, Arnis Parsovs, PhD [41], and independently in [42]. Also, other RSA vulnerabilities were discovered in [43] which justified aborting RSA as an algorithm, not just the company.

According to [1], this was a global crisis for the company which was sued in many other countries. Spain and Slovakia [44] replaced all the physical cards, while Estonia fixed them remotely; then changed the company to **IDEMIA**<sup>14</sup> [45] and, as recommended in [40], moved to threshold cryptography and homomorphic encryption. Currently, the Estonian i-voting system IVXV, [46], also uses **384-bit Elliptic Curve** Cryptography ECC with El-Gamal Encryption (section 4); yet, the list of authorized votes is still signed using 2048-bit RSA key, maybe that's why [47] nullifies the effect of eID problems<sup>15</sup> on IVXV. Note that both IVXV and all digital IDs will have to migrate into post-quantum cryptographic algorithms on the near future.

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<sup>13</sup> Including codes printed too dark that they were readable using torch, without opening envelope (happened in 2002 with the old company then again in 2018: <https://news.err.ee/886313/new-id-card-issue-codes-can-be-read-using-torch-without-opening-envelope>), duplicate email addresses in certificates, issuing certificates with incorrectly encoded public keys, failing to revoke certificates of deceased persons.

<sup>14</sup> **IDEMIA** contract with Estonia will end on 16<sup>th</sup> Nov 2025; Thales is scheduled to take over as the manufacturer of Estonian ID cards (<https://www.id.ee/en/article/thales-id-card/>).

<sup>15</sup> Despite that, we chose to include this subsection in the paper to enlighten the reader that digital identity problems are part of the issues involved in the more general process of i-voting and hence digital democracy. A demonstration of this fact is that when **IDEMIA** started a transition into a new chip platform, Cosmo X, on July 2025 (<http://www.id.ee/en/article/important-changes-with-the-current-estonian-card-issuer-idemia/>) this necessitated some updates in IVXV 2025 version as will be discussed in section 6.



### 3.2 Estonian i-voting before IVXV

As a preface, this section gives a condensed brief on how the Estonian i-voting system has evolved from 2005 to its final form as IVXV.

The main design theme since 2005 is a double envelope protocol sending voter signed ballot after first encrypted by the election public key<sup>16</sup> to the vote collector. Then, we mark 2 milestone step transitions [sec.1 of 48,49,50]:

In 2011, a student named *Paavo Pihelgas*<sup>17</sup> demonstrated a proof-of concept ballot-manipulating software that relied on the absence of an acknowledgement to the voter that his/her vote was received. Hence, [51], the ability for voters to verify their votes was first introduced in 2013. However, several flaws were discovered in 2014-2016, [52], that could maliciously alter the vote or the QR code.

Then, in 2017, *Cybernetica* partnered with *Smartmatic* to produce a new design, IVXV [53], with the QR verification code in its current form. A lot of improvements on IVXV since 2017 include a vote-registration service to guarantee no vote dropping, a shuffling re-encryption mix-net that cryptographically shuffle anonymized vote ballots to guarantee vote privacy, and a *Schnor* based non-interactive zero-knowledge proofs (NIZKPs) to prove the shuffled mix-net output decrypts to the same value as its original input; all of this will be detailed in the next section.

## 4 IVXV

In this section, we explain the design and structure of the Estonian internet voting system, IVXV, as described in the official documents [54]. Then we detail an important cryptographic attack along with its fix before the 2023 elections.

### 4.1 Brief Factsheet

The developing companies are *Cybernetica-Smartmatic* [3,54]; the voting device had to be a desktop PC, smartphone voting was only possible in October 2025 local elections [55]; voting can be done using *mobile-ID*, *Smart-ID*, or any digital identity integrated in the *web-eID*<sup>18</sup>; multiple voting is allowed to avoid coercion or vote

<sup>16</sup> That's why it is called "double envelope"; the vote is embraced with 2 cryptographic keys: the voter key and the election key.

<sup>17</sup> The student filed a complaint, [49], to the Estonian Supreme court requesting to nullify internet votes in 2011 elections. The complaint was dismissed for passing the 3 days limit (<https://www.riigikohus.ee/en/constitutional-judgment-3-4-1-4-11>); an older example that accumulated to our impression on frequently rejecting complains based on deadline. We stress that the authors of this paper have never visited Estonia and do not have any connections with Estonian people except those we contacted as part of the Systemization of knowledge process. **relate this to my largescale attack saying that QR is not enough fix**

<sup>18</sup> The IVXV version for EP-2024 included extra *web-eID assistance service*, *Smart-ID assistance service*,...to scale horizontally enabling the usage of different digital identities (section 2 of the architecture, sections 8.5-8.6 of the protocols in [59]). The web-eID solution (<https://www.id.ee/en/article/web-eid/>, <https://github.com/web-eid/web-eid-system->

buying (only last vote is counted and a poll station vote overrides all i-votes); **El-Gamal Homomorphic** Encryption scheme is used to encrypt votes then the encrypted vote is digitally signed by the voter (double envelope); optional vote verification can be done by voters (through *QR codes* using a second mobile device) within 30 mins of voting with a max of 3 times; a **single re-Encryption Mixnet server** is used to scramble votes before decryption to preserve ballot secrecy. The setup phase, before the election starts, constructs the *election secret key* from 9 parts issued by the members of the *Election Commission of the Republic*, such that decryption requires 5 out of 9 parts<sup>19</sup>. Finally, after the election, there is **an auditor application** (could be run by anyone) that verifies the cryptographic NIZKP proofs provided by IVXV on the election published output data.

## 4.2 System Architecture & Voting Steps

The system architecture and voting steps are depicted in Fig.3, which could be summarized as follows

1. The voter installs the voting application, sometimes abbreviated as **VA**, on his/her PC.
2. After submitting the digital identity ID, the voting application sends to *the vote collector*, **VC**, which in turn sends to *the registration service*, **RS**, to check the eligibility of the voter to vote; if eligible replies with the candidate choices for that voter (according to district) to be displayed to the voter.
3. The voting application encrypts the voter choice using the election public key (El-Gamal encryption), adds the user signature on the encrypted vote (with the voting application running on the voter's PC and after the voter's approval, *the voting application has the right to sign a message with the voter signature*), adds also the signed *timestamp certificate*<sup>20</sup> received from the registration application through the vote collector *after verifying the signatures of both*, and then sends the double envelope ballot to the vote collector.
4. The vote collector application validates the voter's signature; after validation, the signature is removed, and the encrypted vote is added to the list of votes

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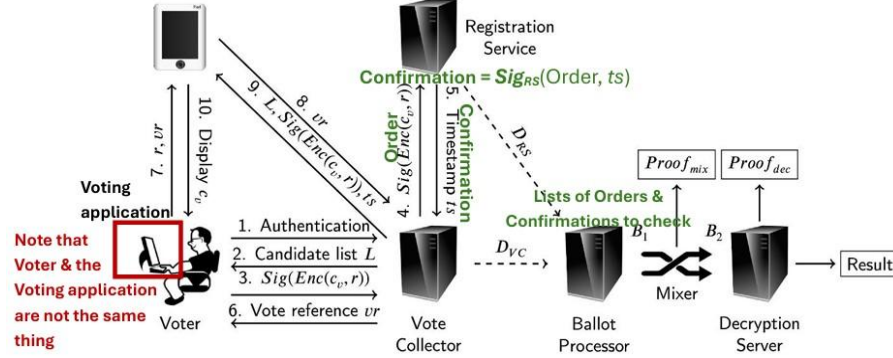
architecture-doc) is part of the European Union web-eID project for all public key cryptography digital identities across Europe.

<sup>19</sup> Hence comes the term “**threshold cryptography**” in the end of section 3.1, because a threshold is agreed upon from the beginning (here 5 out of 9). Relating to the key parts concerns mentioned in section 2, cryptographically, as long as malicious committee members (or leaked key parts) are  $\leq 4$ , they cannot collude to leak the key. Note however, that this does not prevent the key from being leaked through a compromised storage media like the desktop of Fig.1 for example, but the system states this mistake was never repeated and proper cautions are taken since then.

<sup>20</sup> When [1] was written, the timestamp certificate was used only to distinguish the last vote and to check the 30 mins verify duration. In 2024 version, we will see why in section 5.3, the certificate sent by RS to VC is a **signed CONFIRMATION** that contains the original request (**ORDER**) sent and signed by the VC, along with the timestamp.

stored in the *Ballot Processor*. After voting is closed, and before sending ballots to the mix-nets, the ballot processor performs some integrity checks, removes multiple votes and votes overridden by poll station voting. Finally, the remaining list of “to be counted votes” goes through **shuffling** mix-nets (one *Verificatum* server)<sup>21</sup> to hide their original order (in local elections each district votes must be mixed separately as they will be counted separately), then verifiably decrypted at the counting phase.

5. The vote collector sends a verifying **QR code**<sup>22</sup> to the voter for optional vote verifying (through verification application) using a second smart device.



**Fig.3.** A diagram describing the architecture & the steps of the Estonian voting system; adopted from [1] with adding some remarks in red, and new updates in IVXV 2024 in green

### 4.3 Cryptographic Details of Last Fixed Attack

The exploit introduced in 2022 by [49], and fixed in 2023 by IVXV, give us a closer look into the cryptographic details of El-Gamal encryption algorithm; especially since we will mostly treat it as a black box in the rest of the paper.

-Let the election public key be " $y$ " with corresponding secret key " $S_k$ ", and " $g$ " be the generator for El-Gamal encryption over the finite field  $G$ ; hence, the equation «  $y = g^{S_k}$  »<sup>23</sup> holds.

-To encrypt a vote " $v$ " the voting application generates a random number " $r$ ", so that the encrypted vote is  $(C_1, C_2) = (g^r, y^r v)$ <sup>24</sup>

<sup>21</sup> IVXV uses *Douglas Wikström's Verificatum* (<https://www.verificatum.org/>); the package provides a verification application, and IVXV too (and several other projects [48]).

<sup>22</sup> According to [12], there was a revealing incident of *the president vote* through his QR code: he showed it in front of cameras to encourage citizens, and someone took a snapshot. As mentioned in [9], this is not considered a privacy exploit since the President voluntarily revealed his QR-code.

<sup>23</sup> All the presented equations are modular arithmetic over the finite field  $G$ , and in the IVXV used elliptic curve **SecP384bR1**

-The verification application, working instantly within 30 mins, receives " $r$ " from the voting application (hidden in the QR code) and calculates  $v = C_2 / y^r$  where the voter is assured when the displayed " $v$ " is the same " $v$ " he/she voted for.

-When counting votes, the election authority uses the election secret key ( $S_k$ ) and El-Gamal encryption known equation «  $y = g^{S_k}$  » to calculate  $v = C_2 / ((C_1)^{S_k})$

-In the older design, the verification application only receives  $C_2$  from the vote collector. This gives a malicious voting application the chance to manipulate the encrypted cipher text by sending different values of  $C_1$  for the same  $C_2$ . Without checking  $C_1$  value, the verification application will not detect a problem/error if the voting application sent a wrong " $r$ " value to the vote collector; either to drop the vote from being counted (becomes invalid), or to craft a specific  $r'$  such that  $y^r v = y^{r'} v'$  to deceive the vote collector into recording  $v'$  as the voter's intended vote.

The authors found three possible manipulations all with a simple fix (only for verifying voters): making the vote collector send the whole encrypted pair ( $C_1, C_2$ ) to the verification application which should also verify that  $C_1 = g^r$  as was finally done [56] on 23<sup>rd</sup> Feb 2023 just before March 2023 elections. The check methodology has changed in the 2025 version to checking that the pair ( $C_1, C_2$ ) represents a valid point on the election elliptic curve<sup>25</sup>; in the designated attack example; this will echo the invalidity error to the QR code if the adversary tried to drop the vote, or inform the voter that his/her vote was recorded differently ( $v'$  instead of  $v$ ).

In 2022, the authors found it concerning [49/sec. 3.6] that such a straightforward vulnerability wasn't noticed earlier; they also criticized IVXV in general [42/sec. 4] especially in keeping the source code of the voting application secret. It turned out that the Estonian e-voting regulation law [57/§ 48<sup>8</sup>/(7)] clearly states that "The source code for the voter application is not published"; what was promoted recently (30<sup>th</sup> Sep 2025) as publishing the voter application refers to publishing only the new server-side code [58] that receives the already encrypted and signed ballot (step 3 in Fig.3) instead of communicating directly with the vote collector. The PhD in [47] says it is generally considered a national security matter; hence, we add an honest official voting application as a trust assumption in IVXV, otherwise all risks associated with using a malicious voting application remain possible for the official one as long as it is not open sourced.

<sup>24</sup> Now it is clear why a voter revoting with the same choice should lead to a different ballot (and the voter will sign a different cipher leading to a different signature value also); hence, we believe the duplicate vote problem in Fig.2b reflects more than that.

<sup>25</sup> The files in [56] now, since 30/90/2025, contains only a RPC (Remote Procedure Call) *RPC.Verify*; the real check is now done in (<https://github.com/valimised/ivxv/blob/published/common/java/src/main/java/ee/ivxv/common/crypto/CorrectnessUtil.java>) and if an error is detected (not a valid point) it will be seen by all verification applications.

## 5 Updates and Fixes in the 2024 version

In addition to extending IVXV to support voting with more kinds of digital identities like web-id and mobile-id, [59], the IVXV version 1.9.10-EP2024 included many other improvements. This section discusses three important fixes that were committed to the official code site on 30<sup>th</sup> May 2024 before the European Parliamentary elections on 3<sup>rd</sup> of June 2024<sup>26</sup> and the research efforts that led to them; it also discusses providing a fingerprint authentication check (section 5.4) for the voting application. See Table.1 in the Appendix for a summary.

### 5.1 Decryption of Invalid Votes

Invalid votes are now thrown in a separate file and **ZKPs (Zero Knowledge proofs) are generated for correct decryption of invalid votes as well.** However, election observers are not allowed to access this file or verify those proofs and there were a lot of debating and complaints about that (section 2); the issue of observers' rights persists in OSCE 2025 report [10/sec. 3.3/65].

#### *Why not reveal invalid votes?*

The reason is better explained scientifically as T. Kraavi did in his master thesis, [18], than by the state election service as recorded in the supreme court decision [15].

-Reasons 1&2 in [15] talk about the technical infeasibility of decrypting invalid votes after the election and how this needs parts of the secret election key. The fact is, this was already done; meaning it is feasible, and the statement was kind of *misleading the court*. The interested reader can find the scientific details in [18] and/or trace the code in ***DecryptTool.java*** file [60].

-While the 3<sup>rd</sup> reason in [15] of “*not knowing in advance what the invalid ballot contains and it may be an attack*” seems vague and not convincing, the argument is rationalized in [18]; it is the possible reveal of some information about the voter of the invalid vote, or more severe the threat of ***encoding attacks*** described in [61/ sec. 3.3 & 4.1] where an adversary can know the votes of several voters if able to submit a carefully crafted invalid vote and also view its decryption. Hence, the rational is to shrink the circle of trust into auditors only, which is almost not needed if invalid votes were rejected earlier by the system (done in 2025 as will be discussed in section 6.). Note, however, that voters can make their own voting application that allows invalid votes, and accompany it with a verification tool that enables them to view the vote decryption [21,22]; this can defeat both cautionary steps and make the system vulnerable to encoding attacks if done by an insider. In fact, tracing the ***number of invalid votes*** in the official statistical site [4] to be ***exactly 1*** in three local elections since 2021, **increased to 2 in 2024 then 3 in 2025 with the issue being further discussed**, makes it look quite suspicious; the doubt includes anyone who can see the votes.

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<sup>26</sup> Note that the time difference here is 3 days, while it was nearly 3 weeks in 2025; this is related to the delay complaint in [23,24].

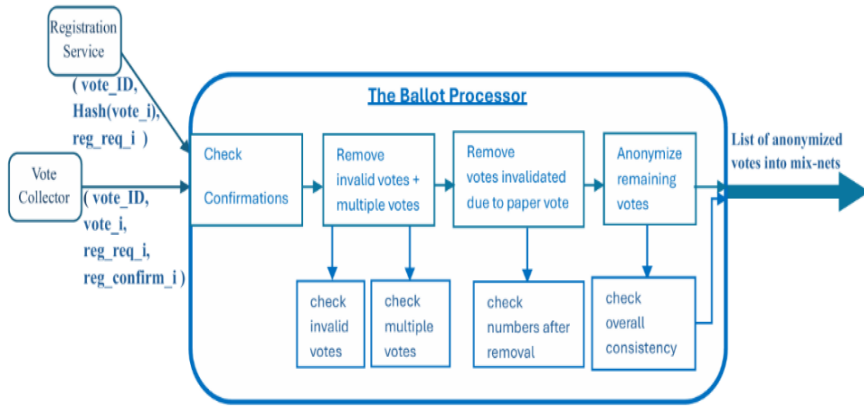
## 5.2 Integrity Checks

Another problem that was mentioned in OSCE 2023 report [2] is that the authorities are assumed trusted regarding the deletion of multiple votes or ill-formed vote ballots. Quoting their own words "The critical step of removing the votes overwritten by another vote cast on the internet or in a polling station was not audited", "An insider with sufficient resources to alter the system, if able to do so undetected, could manage to control which votes are removed and therefore partially impact the results". This was viewed by [62] as trusting the vote collector (VC) and registration applications (RS) to not collude, otherwise it would be possible to drop ballots; the *Ballot Processor* in Fig.3 could also manipulate the ballots (assumed trusted by the system). To elaborate more, yes there are decryption proofs that what goes into the *mix-nets* is exactly what gets out of it to be finally decrypted, and yes there is the possibility to design a public independent decryption proof verifier [48], but there was no cryptographic proof for the transition from the total list of votes to the "to be counted" list of votes; what is called the *processing stage* and we believe is part of *universal verifiability*.

### IVXV Adopted Solution

What we believe is partial protection from this vulnerability (see section ??), was integrated into the audit application of IVXV through the files, [63], *Integritytool.java & Intcheck.py* in May 2024; the details of the solution were published academically in [64] as of 23/12/2024. The authors state that they have contacted IVXV team with their proposed checks to detect insider risks at the processing stage, and that it was successfully deployed.

The proposed checks are applied to the Ballot Processor data which is an offline computer [64/sec. 3.A] that performs some processing on this data (step 4 in section 4.2) to then input the list of anonymized votes to the mix-nets as shown in Fig.4.



**Fig.4.** The (modified) processing stage after adding the *integritytool.java* file

Since the votes file (along with the necessary checksums) is cryptographically signed after each step, most examinations depend on comparing SHA256 hashes of subtotals,

totals, and individual votes before and after each step. Also, *count-based validation* was needed to detect certain types of manipulations like adding the removed older multiple votes to the list of anonymized votes; i.e.,

$$\boxed{\text{Count (original votes file)} - \text{Count (anonymous valid votes)} = \text{Count (multiple votes)} + \text{Count (replaced by paper)} + \text{Count (invalid votes)}} \dots \text{Eq(1)}$$

Note that the Ballot Processor receives the list of all votes along with any necessary confirmation checks [59/protocols.pdf/Fig.6.3]; i.e., the integrity of those cryptographic checks (like the *Count (original votes file)*) depends on trusting the vote collector and the registration application to not collude [62]. Hence, what was mitigated by [63,64] is **the risk of a colluding Ballot Processor**; recall that it also helped in discovering other technical errors like those in Fig.2 of section 2.3.

### 5.3 Session ID & Time Checks

A possible attack by a malicious voting application that could deceive even verifying voters was discovered by Olivier Pereira in [50]; a malicious voting application could fake a system crash to deceive the voter to vote again. By doing so, the application can take the voter signature twice (generate another "*r*" value to construct and sign a new encrypted vote in the background); hence while showing the voter the QR code of his/her choice, the system will consider it an old vote and will use the new vote.

#### *Previous Suggestions*

- Olivier Pereira suggested a few mitigations in his paper, K. Krist's PhD [47/sec. 5.3] suggested others; none was actually adopted, and we will mention them as we go.

1. One may also recommend advising voters to double check the number of voting transactions with other available e-government services available in Estonia like *myID* service [65], especially if their device has suffered a system crash while voting.
2. Another simple safeguard from this specific problem, [66], would be *to force a time interval between votes*; the verification interval, *30 mins*, seems a suitable choice. However, this must be accompanied by heavily warning voters to close the application then reopen again; if the voter eID remained available on the voter's PC more than 30 mins, a *Ghost Click attack* becomes possible [50] and a malicious voting application would have enough time to submit without voter's knowledge.
3. Both Pereira [50] and [67] suggested allowing to verify the last vote only, we thought of adding a flag to the QR code on whether this is the last vote or not.
4. While adding a timestamp to the QR code won't help the voter<sup>27</sup>, a voting attempt tracking number as suggested by [50,67] will change in the revoting screen after the faked crash. Specifically, [67], the voting application should

<sup>27</sup> Even if the voter noticed the small-time difference, could think "maybe the system records the time when I started, not when I finished"; a vote counter can't reveal following votes either.

generate a session tracking number that is shown on the voting screen, added to the ballot, and checked to be new each time; the number should be embedded in the QR-code for the voter to check that it equals the corresponding number on the voting screen. However, some may view asking the voter to trace numbers as decreased usability.

-Jan Willemson argues in [9/sec. 4] that all such solutions could be used by a coercer/vote buyer to find out if the voter voted again, we believe a coerced voter is only at this risk for 30 mins (the verify duration); besides such risk exists anyway through services like *myID*, [68,69], as will be discussed shortly.

#### ***Formal Verification & Extended Pereira Attack***

Published on June 2024, [67] introduced for the first time an automated formal verification of IVXV security (treating cryptographic functions as black boxes). Their tool rediscovered the Pereira attack and another *timing attack* that we believe is a generalization of *Ghost click* attack mentioned in [47/ref.166, 50]; the adversary stores the fake vote to submit it as late as possible to guarantee its count. We believe IVXV deployed the time difference & session ID checks suggested in [67].

#### ***IVXV Adopted Solution***

A solution to the generalized attack in [67] was added on 30<sup>th</sup> May 2024. The code was updated to check the session-ID is still the same before generating the verifying QR; the code documentation [70/lines 22&105&167]<sup>28</sup> highlights that this “**prevents reusing session ID until it is deleted from a database or expired**”.

Tracing a little deeper, [59/protocols PDF/sec. 8], the *PKIX* (*Public Key Infrastructure X.509*)<sup>29</sup> timestamp protocol is used by the registration service to record the time of casting the vote, while the *rsyslog* service records the logging time in milliseconds which make it possible to use the *Guardtime* module to ensure the integrity of the logs.

### **5.4 Authenticating the Voter Application and the Pereira Attack**

As [67] warns, the solution adopted in section 5.3 cannot handle the original attack in [50], since it depends on the fast sequencing of the fake crash-restart scenario; this allows a malicious voting application to use the same session ID twice and have a reasonable time difference.

According to our study of available resources, IVXV published in 2024 the fingerprint of the official voter application with a documentary, [71], guiding voters

<sup>28</sup> The file has slightly changed in the new version released 30/9/2025; checking the session-ID and the time limit for a voting session is in the server-side of the voting application, [58], and almost every code that deals with the vote.

<sup>29</sup> *PKIX* is a timestamping protocol that enables a trusted third party (called *TMS* in [66]) to confirm the existence of a specific data at a specific point in time with its signature; can check 4 times *thisUpdate*, *nextUpdate*, *producedAt*, *revocationTime* (<https://datatracker.ietf.org/doc/html/rfc6960>, [https://link.springer.com/referenceworkentry/10.1007/0-387-23483-7\\_302](https://link.springer.com/referenceworkentry/10.1007/0-387-23483-7_302))



on how to check it after the download. Other than that, they sufficed with the session ID check up till now in 2025 version; they adopted the argument in [9/ sec 4] that diminishes the effect of the attack to  $(1-p)^n$  for  $n$  voters where  $p$  is the probability of a voter reporting a crash when it happens.

However, we believe that the effectiveness of reporting the crash, and authenticating the official voter application to begin with, holds as an argument only **if it is assumed trusted**. Removing the trust assumption necessitates publishing the source code of the voter application (VA) for voters (and possibly observers) to review the code and then compares its digest to that of the downloaded one; otherwise, if the probability of a malicious official VA is not negligible, reporting a crash would be like reporting a crime to one of the suspects.

## 6 Improvements in the 2025 election

In addition to publishing the server-side part of the voting application, [58], and the earlier publishing of the election source code in general, IVXV 2025 version 1.10, [72], included many other updates that were committed to the official code site around 30<sup>th</sup> Sep 2025 before the local elections on 19<sup>th</sup> of Oct 2025. This section covers technical implementation updates like changing the ballot vote format and the way El-Gamal encryption is verified, then moves to discuss in detail the significant improvements of using Range Proofs to prevent invalid votes from entering the system. These improvements are also included in Table.1 of the Appendix.

### 6.1 Technical Enhancements

The updates in include, as detected by the authors of this paper, but maybe not limited to:

- Changing the ballot format, [58], in a better way to include the checksums integrity checks needed in section 5.2 and dynamic qualifying services (like the voter eligibility to vote or any other future purposes); this was reflected in the extra effort [21,22] needed to adjust their private VA, and the first error in Fig. (2-a) also came from it.
- The dynamic qualifying services in [58] includes also a check of the voting duration time to enforce a tighter time limit on the session ID; i.e., further enhancement of the session ID tracking safeguard discussed in section 5.3.
- Some changes to handle the new eID chip platform discussed in footnote 15; one can also trace future plans (TODO parts of code) for the upcoming new Thales cards (footnote 14).
- Switching the regular check of El-Gamal encryption pair (the vote), which involves exponent raising, to the more efficient check of it constituting a valid point on the election elliptic curve **SecP384bR1**, [73], as discussed in section 4.3
- **Enhancing the mix-net shuffling** code, [74], to add support for the elliptic curve cryptography methodology introduced above. We could not accurately

trace repeating the mixing into more extra rounds as previously criticized; however, the new protocol file [72/protocols.pdf/secs 7.1,10.1] explains that in this local election votes are first grouped according to district before mixing.

## 6.2 Range Proofs

To eliminate any complaints (like [15]) and/or risks (ex. [61]) accompanying invalid votes, it is much safer to prevent them from reaching the Ballot processor at all. The thesis in [18], followed by the scientific report [75], suggested the use of a Zero Knowledge Proof (**ZKP**) check by the Vote collector application to check the validity of the vote it receives from the voting application without revealing it and hence reject invalid votes before being added to the list of votes.

-The thesis preferred the use of **Range Proofs** as opposed to **Set-Membership proofs** for their simplicity and suggested some mitigations to the discontinuity of the set of vote choices. The authors proposed Range Proofs that are based on **Bullet Proofs & Pedersen Commitments**, since they depend on the Discrete Logarithm problem like El-Gamal encryption used in IVXV. They considered general purpose SNARKs (Succinct Non-Interactive ARgument of Knowledge) based on polynomial commitments not suitable for El-Gamal based voting systems. The interested reader can find more discussion of this ZK specific choice on the extended version of this paper [76]; the Range proofs presented above are what is implemented in the 2025 IVXV version (with TODO parts to optimize proof aggregation for larger ballots). The validity check can be clearly traced in the published server-side part of the voting application, [58], which performs different checks (including Range Proofs) on the vote before sending it to the vote collector. The Range Proof check was also added to the verification code because a private voting application can still bypass the code in [58] and send votes (possibly invalid) to the vote collector directly; this fact was proven through the official statistics, [4], showing 3 invalid votes in the 2025 elections (probably the 3 votes in Fig.2). We believe the risk could be eliminated completely if range proofs were deployed through the vote collector to check even votes from other private voting applications.

## 7 Remaining Issues & Further Research suggestions

This section collects existing research that either points to remaining vulnerabilities in the IVXV Estonian i-voting system or suggests further enhancements to strengths its security, including the Estonian efforts for future post-quantum migration, then seals with a holistic view of the status quo. Table.2 in the Appendix summarizes the findings of this section.

## 7.1 Adding NIZKP (online) to remove trust assumptions

This was first suggested in 2022 by [61]; to add a *NIZKP of knowledge* to each encrypted vote as a protection from privacy attacks. The 2024 version of IVXV (1.9.10) partially responded; the Registration Service now sends *hash(vote)* to the Ballot Processor (Figs. 3&4) as explained in section 5.2. Tracing the code [63/line 239], vote hashes are stored in a *Treebag* which represents a binary search tree data structure in Java; i.e., even if it reflected a Merkle tree design, the number of nodes in a Merkle tree (votes) is not cryptographically verified. However, it is not just privacy attacks, and it is not just the Ballot Processor that we need to remove trust from.

In 2023 [62] acknowledged the risk if the Vote Collector (VC) and the Registration Service (RS) colluded together and the 2024 [67] usage of automated formal verification methods [77] asserted the same result; when analyzing IVXV End-to-End-Verifiability (EEV) and privacy under 9 different combinations of trust assumptions [67/Table1] for an adversary controlling more than “some voters’ credentials”,  $EEV^+$  (that maps to current IVXV status) fails if 2 of (VC, RS, timestamping service TMS) colluded. Also, both the OSCE/ODIHR 2023 [2] and 2025 [9/sec. 3.3/28] reports mentioned persisting political parties concerns regarding insiders’ trust.

Hence, the 2025 E-Vote-ID conference contained two suggestions of using Vector Commitments (usually referred to as Verkle Trees); the first [78] is a poster that suggests the use of Verkle trees based on KZG commitments, and the second is a paper [79] suggesting those based on Pedersen commitments. Readers interested in deep cryptographic analysis of the differences can find it in [80]; however, we anticipate that the next version of IVXV will deploy Pedersen based vector commitments. The suggestion in [79] aligns with the system’s previous choice in Range Proofs, the authors are from the IVXV team, and they claim faster simulation time of less than 1 sec; in fact, we detected (with the help of AI) a fork in the current code where they have already started working on it. [Double check this fork issue](#)

## 7.2 Integrity-Coercion tradeoff and other existing sources of information

The PhD in [47] was the first resource we encountered, 2022, concerned with the tradeoff between integrity vs coercion resistance when discussing mitigations for handling a corrupted voter device. Then, in 2023 M Poder (the same observer in [12,13]) presented it clearly in a poster [68] how the *myID* service [65] can provide hard copy evidence of the last e-vote in Estonia which destroys the multiple voting defense against vote coercers/buyers (recall the discussion in section 5.3).

Hence, in an earlier version of this paper, we raised the broader question of *to what limit can the information provided by general purpose activity logs of digital identities* (in any country that uses digital identities in online voting) *help vote buyers/coercers* in catching voters who tries to deceive them. Then, in Oct 2025 [69] discussed the specific Estonian case and suggested different possible solutions like making election transactions secret or obfuscated or even using a special ID card for e-voting that doesn’t allow such integrity checks on its transactions.

A final remark on this context is that suggestions that involve applying aggregating queries as consistency checks on the election results numbers, like [78], can still be deployed with any of the suggestions in [69]. Those queries are assumed to use zero knowledge techniques; ie., they do not necessitate revealing the raw transactions data.

### 7.3 Automated Formal Verification and Mix-net Servers

Although formal verification tools in general, like [81,82], treat cryptographic details as black boxes that one may diminish their value in testing IVXV, two independent attempts were introduced in June 2024. The first, [67], was discussed in sections 5.3 and 7.2; the second [83] was mainly concerned about mix-net servers. The authors used *ProVerif*, [82], to discover [83/ sec 4.3 & Table 5] the same privacy attacks in [61] showing that they can only be defeated using a strong NIZKP for each vote (section 7.1); the paper also contains a comparison analysis between exponentiation mix-nets and re-encryption (like IVXV) mix-nets for the interested reader.

It is also worth discussing that IVXV uses only one single mix-net server<sup>30</sup>. Although Verificatum uses in its documentation, [48/ref.1], an example of 3 mix-net servers, IVXV uses its own implementation of one single server; the argument is simplicity and that the distributed key parts provides an additional security layer. As a basis of comparison, the Swiss-Post i-voting system used in Switzerland has 4 mix-net servers and [84] discovered that NIZK proofs must be verified after each server not just at the end of the whole shuffling process.

### 7.4 The Voting/Voter Application (VA)

We believe the voting application is the most persisting and/or serious integrity problem in the Estonian i-voting system. All risks associated by previous literature with a corrupted voter device, or a corrupted communication channel, are perfectly feasible with a malicious voting application and even more.

In addition to the usability issues regarding the available authentication method, [71], of the official VA; as long as the code is not published, [57], it is based on a trust assumption of it not being malicious. Yes, QR code verification is enough for *individual verifiability*, but with more than 90% non-verifying it is not enough evidence for universal verifiability; the election result may be very well altered by a malicious voting application. In fact, anyone who has enough access rights to edit a few lines of the secret code, or upload a different file to the site, can do that. If non-verifying voters can be identified somehow, for example through social engineering, traditional techniques for Public Bulletin Board (PBB) attacks [85] can be used here. In addition, the current situation leaves it as an open challenge for adversaries to design a malicious VA that can invoke any hidden undiscovered exploits to deceive even verifying voters.

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<sup>30</sup> We thank a previous anonymous reviewer of an earlier version of this paper for pin pointing that weakness point.

Although [9] argues to diminish the effect of the attack in [50] along with the threats in [2], new technologies like online coding through blockchains and smart contracts, [86], have introduced more capabilities to render automated large-scale attacks and/or execution attests for vote buyers/coercers, [37], feasible.

The suggested solutions are mostly straight forward; change the law to publish the official voting application with a more simplified batched authentication [87] to check the published code is the used one; allow only private VAs that are pre-registered and hence examined for malicious code before the election; encourage voters to verify; alarm voters if their vote was casted using a private VA; also, connected to the alarm, encourage voters to vote in the first place (maybe by adding a “reject all” or “boycott” option) because the risks can be magnified for absent voters as there is no QR code to check (recall [36] from section 2).

Stopped here 10/11/2025 9:29 am, maybe have to update the figure too with a different color (green refers to 2024 updates, say blue or magenta for 2025 updates) Still have to check whether there are invalid vote in 2025.

We traced many improvements in IVXV 1.9.10-EP2024<sup>31</sup>, and then 1.10-E2025 meaning we could have missed some. However, we know the newest version is the one analyzed in [38,39] which identified 6 threats with risk level higher than small; the report was done in collaboration with the election authorities (i.e. not biased against or missing updates).

Another general problem is that IVXV neither welcomes communication nor advertises its progress to researchers; the authors of [49] were not informed of the fix in section 4.3 till their paper was published; the authors of [67] seem to lack any contact with IVXV and had to work on earlier versions. The improvements in section 5 were discovered through repeatedly scanning the academic literature and tracing the code; the **existing** English documentation files still holds the 2022 version; we had to dig our way through browsers translators in 2024 then again in 2025.

## 7.5 Post Quantum Cryptography (PQC)

In general, the retaliation game for political elections is more severe than that of financial systems; some countries are willing to pay much more to break a political election than a thief wanting to steal money. In addition, there is the ***Harvest Now Decrypt Later (HNDL)*** threat of storing encrypted data to decrypt them later when QC is possible [57]. Hence, all countries deploying e-voting/e-government systems should be more motivated, without rushing unstudied, to provide post-quantum solutions; in fact, e-governments/e-IDs would be a probable main target of a quantum adversary than e-voting systems as stated by most interviewed experts in [75].

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<sup>31</sup> GitHub shows 897 changed files with 34,059 additions & 10,830 deletions. Translating [59], a lot of work was done in integrating different kinds of IDs and in coordinating with XRoad service (X-tree). A whole section is dedicated to Registration Service, [59/protocols/sec.6]; the interaction between online (RIA), offline (RVT) and other IVXV services.

### ***Estonian Quantum Efforts***

For our specific case of Estonia, the authors of [75] included 5 out of 24 experts from Estonia in their interviews, and covered steps taken by the Estonian government [75/sec. 2.2.2] in different quantum related aspects.

-A European cooperation to deploy a quantum communication infrastructure in 2020 [80] led to the NordIQEst (*Nordic-Estonian Quantum Computing e-Infrastructure Quest*) in 2022 [81]

-A collaboration between *Cybernetica* and University of Tartu to create post quantum solutions has started in 2021 [82] including jointly supervised doctoral thesis from which we encountered a paper (2024) on PQC migration obstacles in Estonia [67].

-Finally, PQC is 1 of 6 challenge areas included the *Cyber Security hub* established in 2023 between Estonia and a major Czech ICT powerhouse both in industry and education [83].

Then, the European Commission published on 11 April 2024, [84], a coordinated PQC implementation roadmap for all member countries. The announcement contained only one guiding statement; the implementation should be “*via hybrid schemes that may combine Post-Quantum Cryptography with existing cryptographic approaches or with Quantum Key Distribution (QKD)*”. This was different from ENISA 2021 report [70/page 35] which excluded solutions using *QKD* because its setup relies on pre-established authenticated communications channels; [75,85] also reported different experts’ opinions on this point. In fact, [67] considered “*EuroQCI focused attention on QKD technology*” as an obstacle to post quantum transition since they believe its functionality is limited compared to PQC.

### ***Estonian I-Voting Related Efforts***

Digital IDs are a crucial element in i-voting; the *ENISA* July 2024 report [86] on the unified European digital identity project *eIDAS*, though, did not specify the PQC transition details. Then on 19 Feb 2025, [87], *Cybernetica* announced that the first hardware chip enabling post-quantum ID card has been certified in Europe.

When asked about post quantum i-voting, they answered “*there is more math to do*”. In their 2024 paper [67/sec. 2.E], they considered the complexity of managing multiple layers a disadvantage in most existing Hybrid systems. For i-voting, [66/sec. 4.E], this would be a problem with the needed multiple layers to shuffle votes in *mix-nets*; we think lattice-based approaches are advantageous here, [85], as they depend on matrix operations. The extended version, [65], contains a brief survey on post quantum research on i-voting and replacements for El-Gamal encryption.

## **8 Possible Research Directions**

In this section we go through possible research threads that could evolve from the gathered material, even if not necessarily pursued by the authors of this paper.

The work in [59] could be a starting point for researchers concerned with hardware acceleration of Zero Knowledge Proofs that goes in depth into circuit specific details,

while [60] would be useful for researchers interested in theoretical cryptographic proofs. Also, research can be further pursued to conduct comparative analysis between possible zero knowledge proofs concerning implementation details along proof batching reductions over the exact 384-bit elliptic curve used in IVXV.

However, one may think the future will promote more research on efficient quantum secure solutions and less encouragement to elliptic curve research. The research space is challenging, [69], on the efficiency-security tradeoff; demanding for new innovations [71], for cryptanalysis [73,74], and formal verification [85] of the new PQ protocols (how to prove correctness and security if it is not verifiable on existing hardware [57]). In addition, AI-based attacks could be another rich research area; side channel attacks, [74], “*how information can be leaked from the implementation of a cryptographic component*” and maybe from quantum devices too.

Finally, with the intuition of [53], it could be an interesting research to tackle the broader question of *to what limit can the information provided by general purpose activity logs of digital identities* (in any country that uses digital identities in online voting) *help vote buyers/coercers* in catching voters who tries to deceive them, and whether a blockchain based e-government is an advantage or disadvantage in that direction. We have shown in section 6.4 the information *myID* [47] and similar services can provide; to what limit could adversaries use them to defeat, for example, the *CK* (*Complete Knowledge*) authenticity metric [65/footnote 21] whether for vote buying/coercion or any other malicious activity.

## 9 Summary & Conclusions

In this paper we gave a political and technological historical brief on the development and status quo of the Estonian internet voting system. Then we explained the current system architecture and surveyed available material from academic literature and different other resources to cover reported attacks and/or vulnerabilities and how they were fixed by the system.

We addressed the concerns of opposing parties in their meeting with ODHIR [9/sec.3.3/28], and introduced solutions to some vulnerabilities; mainly, authenticating the voting application and cryptographically proving the list of votes against insiders’ manipulation.

We suggested some possible solutions that give voters the freedom to deliberately use a different voting application they trust more; the first is allowing the voter to check the application fingerprint prior to voting; the second is allowing only a list of pre-registered applications (through digital signatures) and informing the voter with the result. In addition, we introduced the threat of *encrypted copy attacks* that could be conducted at large scale, [88,89], using an adversary’s voting application; hence, we highly recommend IVXV to adopt a solution to the voting application problem. We also recommend deploying Range-proofs as suggested by Tallinn University thesis, [17], to get rid of invalid votes problems.

Then, we proposed alternatives for more robust guarantees that the list of votes was not manipulated by insiders; one is to extend the currently deployed consistency

checks and add RLAs comparing ballots to general records of digital transactions interacting with IVXV. The second is aggregating votes (online during the voting process) in an authenticated data structure like Verkle Trees which we detailed as our proposed solution. This will eliminate the possibility of colluding insiders performing the above attack and will contribute to more public trust.

Finally, we highlighted the quantum computing threat and ended by introducing a variety of possible research threads that could evolve from all the paper material.

As a closure, we advise the IVXV team to respond more scientifically and objectively to OSCE, complaints, and researchers; this will give them credit for their unnoticed work and will lead to a better understanding.

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**Appendix A : Summarizing Tables**